Extended Analysis and Reproducibility Companion for "Prescribed-Time Control Beyond Equilibria with Terminal Phase Guarantees"

This companion collects additional analytical refinements and secondary illustrations for the manuscript. All complete proofs of the main results, analytic certificates, and primary numerical definitions are in the manuscript and its Technical Supplement. The material below extends that submitted argument. We retain their notation and bibliography. Unprefixed result and equation references marked main refer to the manuscript; S-prefixed equations belong to the Technical Supplement, and R-prefixed equations and figures belong here. Computational methods C01–C09, Tables C1–C4 and their archived data remain in the existing computational companion at https://anonymous.4open.science/r/Scheduled-network-control-2950/.

I. Additional Analytical Refinements

Retain the deadline $T > 0$, scalar contraction rate $\rho > 0$, deterministic initial error η_0, and measurement-noise intensity σ from main equation (32). Expectation and variance are denoted by $\mathbb{E}$ and Var. Terminal mean-square error (MSE) is $\mathbb{E}[\eta(T)^2]$, and root-mean-square (RMS) error is its square root. The stochastic refinements use $\sigma > 0$; zero-noise cases are identified explicitly. We use $f \sim g$ for $f/g \to 1$, and retain the main paper's O, o, and Θ asymptotic conventions.

A. Ideal uncapped inverse-square variance

For the ideal stochastic equation obtained from main equation (32) by using the uncapped κ_2, write $s_t = (T-t)/T$. With P_ρ defined in Technical Supplement (S1), its variance for $t < T$ satisfies

$$\mathrm{Var}_{\kappa_2}\,\eta(t) = \frac{\rho^2\sigma^2}{T}\left[P_\rho(s_t^{-1}) - e^{-2\rho(s_t^{-1}-1)}P_\rho(1)\right] \sim \frac{\rho\sigma^2}{2T}s_t^{-2}. \tag{R1}$$

Thus ideal white-noise variance grows as s_t^{-2} even though the deterministic error decays. Bounded realization changes the accuracy question to terminal MSE at a finite cap. Main Theorem 24 establishes the resulting optimized inverse-square law.

B. Inverse-square derivative refinements

We refine the noise sensitivity of the inverse-square optimum whose moments, global minimization and Lambert remainder are established in main Theorem 24. Fix $\rho, T > 0$ and $\eta_0 \neq 0$, and let $\sigma \downarrow 0$. Write $\theta_{\mathrm{opt}} = \sqrt{TK_{2,\mathrm{opt}}}$ and $\mathcal{M}_2(\theta) = \mathcal{E}_{\kappa_2,\theta^2/T}$, following the proof in Technical Supplement Section I-D. Set $D_\rho = (2\rho^2 - 2\rho + 1)/(4\rho)$, $x_\sigma = 4\rho\theta_{\mathrm{opt}}$ and $H_2(x) = xe^x + (x-4)e^{x/2}$. With $Z_\sigma = 16\rho T\eta_0^2 e^{2\rho}/\sigma^2$, the exact stationarity equation at the interior optimum is

$$x_\sigma e^{x_\sigma} + (x_\sigma - 4)e^{x_\sigma/2} = Z_\sigma\left(1 - \frac{D_\rho\sigma^2}{T\eta_0^2}\right). \tag{R2}$$

Let W be the principal Lambert function, defined by $W(z)e^{W(z)} = z$ for $z \geq 0$, and put $W_\sigma = W(Z_\sigma)$. Taking logarithms in (R2) and applying the mean-value theorem to $x + \log x$, as in Technical Supplement Section I-D, gives

$$x_\sigma - W_\sigma = O(e^{-W_\sigma/2}) + O(\sigma^2). \tag{R3}$$

At an interior stationary point, (R2) eliminates the terminal-bias term exactly and yields

$$\mathcal{M}_2(\theta_{\mathrm{opt}}) = \frac{\sigma^2}{32\rho T}\left[x_\sigma^2 + 2x_\sigma(1 - e^{-x_\sigma/2})\right]. \tag{R4}$$

Differentiating the exact relation (R2) gives the local noise sensitivity directly. Its right-hand side is $Z_\sigma - 16\rho D_\rho e^{2\rho}$, so

$$\frac{dx_\sigma}{d\log\sigma} = -\frac{2Z_\sigma}{H_2'(x_\sigma)}. \tag{R5}$$

Define $\mathrm{RMS}_{2,\mathrm{opt}} = \sqrt{\mathcal{M}_2(\theta_{\mathrm{opt}})}$, as in main Corollary 25. Set $Q(x) := x^2 + 2x(1 - e^{-x/2})$. The exact derivative $H_2'(x) = (x+1)e^x + (x/2-1)e^{x/2} = (x+1)e^x[1 + O(e^{-x/2})]$ has a quantitative relative remainder as $x \to \infty$. Differentiating the exact identity (R4) using (R5) gives

$$\frac{d\log \mathrm{RMS}_{2,\mathrm{opt}}}{d\log\sigma} = 1 - \frac{Z_\sigma Q'(x_\sigma)}{H_2'(x_\sigma)Q(x_\sigma)} = 1 - \frac{2}{x_\sigma + 2} + O\left(\frac{e^{-x_\sigma/2} + \sigma^2}{x_\sigma}\right).$$

The second equality uses (R2) and $Q'/Q = 2(x+1)[x(x+2)]^{-1}[1+O(e^{-x/2})]$. Equation (R3), $e^{-W_\sigma/2} = O(\sigma\sqrt{W_\sigma})$ and $2/(W+2) - 2/(W+1) = -2/[(W+1)(W+2)]$ then give

$$\frac{d\log \mathrm{RMS}_{2,\mathrm{opt}}}{d\log\sigma} = 1 - \frac{2}{1+W_\sigma} + O(W_\sigma^{-2}). \tag{R6}$$

For the envelope ratio, the exact identity $d\log(\mathrm{RMS}_{\mathrm{env}})/d\log\sigma = T\eta_0^2/(T\eta_0^2+\sigma^2)$ and (R6) give

$$\frac{d\log(\mathrm{RMS}_{2,\mathrm{opt}}/\mathrm{RMS}_{\mathrm{env}})}{d\log\sigma} = -\frac{2}{1+W_\sigma} + O(W_\sigma^{-2}).$$

These local derivatives refine main Corollary 25 with each family evaluated at its own optimal cap. The reference remains the full bounded-gain accuracy envelope. For fixed $\alpha \in (0,1)$, the equality gain in main equation (36) satisfies $\partial_w v_\alpha^\star = -\rho(v_\alpha^\star)^2 < 0$, whereas the capped power profiles are nondecreasing. This distinguishes the decreasing benchmark with finite accumulated gain from the deadline-directed families. Companion C09 contains the scalar fit records.

C. Normalized power-law family

The inverse-square construction is the case $p = 2$ of $\kappa_p(t) = T^{p-1}(T-t)^{-p}$, with fixed $p > 1$. For $K \geq T^{-1}$, define the capped profile $\kappa_{p,K}(t) = \min\{\kappa_p(t), K\}$ and $\theta_{p,K} = (KT)^{1/p} \geq 1$. The crossing time is $t_K = T - T/\theta_{p,K}$, and the total capped gain $\mathcal{J}_{\kappa_p}(K) = \int_0^T \kappa_{p,K}(t)\,dt$ satisfies

$$\mathcal{J}_{\kappa_p}(K) = \frac{p}{p-1}\theta_{p,K}^{p-1} - \frac{1}{p-1}, \qquad K(T-t_K) = \theta_{p,K}^{p-1}. \tag{R7}$$

Indeed, $\int_0^{t_K} \kappa_p(t)dt = (\theta_{p,K}^{p-1}-1)/(p-1)$. For $w < t_K$, the terminal transition exponent is

$$\int_w^T \kappa_{p,K}(u)\,du = \frac{p\theta_{p,K}^{p-1} - (T/(T-w))^{p-1}}{p-1}. \tag{R8}$$

Let $V_{p,\mathrm{pre}}$ and $V_{p,\mathrm{cap}}$ be the variance contributions before and after the gain reaches its bound in main Proposition 20. The substitution $z = T/(T-w)$ gives

$$V_{p,\mathrm{pre}} = \frac{\rho^2\sigma^2}{T} e^{-2\rho p\theta_{p,K}^{p-1}/(p-1)} \times \int_1^{\theta_{p,K}} z^{2p-2} e^{2\rho z^{p-1}/(p-1)}\,dz. \tag{R9}$$

For fixed p, ρ, T, endpoint Laplace expansion as $K \to \infty$ and direct cap integration yield

$$V_{p,\mathrm{pre}} = \frac{\rho\sigma^2}{2T}\theta_{p,K}^p e^{-2\rho\theta_{p,K}^{p-1}}[1+O(\theta_{p,K}^{-(p-1)})], \tag{R10}$$

$$V_{p,\mathrm{cap}} = \frac{\rho\sigma^2}{2T}\theta_{p,K}^p(1-e^{-2\rho\theta_{p,K}^{p-1}}). \tag{R11}$$

Now fix $\eta_0 \neq 0$ and let $\sigma \downarrow 0$. Write $K_{p,\mathrm{opt}}$ for a global MSE-minimizing cap, $\theta_{p,\mathrm{opt}} = (TK_{p,\mathrm{opt}})^{1/p}$, and $\mathrm{RMS}_{p,\mathrm{opt}} = \sqrt{\min_{K\geq T^{-1}} \mathcal{E}_{\kappa_p,K}}$. The leading small-noise objective is

$$H_p e^{-c_p\theta_{p,K}^{p-1}} + \frac{\rho\sigma^2}{2T}\theta_{p,K}^p, \quad H_p = \eta_0^2 e^{2\rho/(p-1)}, \quad c_p = \frac{2\rho p}{p-1}. \tag{R12}$$

Its stationarity relation, together with the exponentially small corrections in (R10)– (R11), gives

$$\theta_{p,\mathrm{opt}} = \left[\frac{W(Z_{p,\sigma})}{2\rho p}\right]^{1/(p-1)}[1+o(1)], \tag{R13}$$

$$Z_{p,\sigma} = 2\rho p\left(\frac{4TH_p}{\sigma^2}\right)^{p-1}. \tag{R14}$$

Since $K = \theta_{p,K}^p/T$ and the leading variance is $\rho\sigma^2\theta_{p,K}^p/(2T)$,

$$K_{p,\mathrm{opt}} = \Theta\left(\log^{p/(p-1)}\frac{1}{\sigma}\right), \qquad \mathrm{RMS}_{p,\mathrm{opt}} = \Theta\left(\sigma\log^{p/[2(p-1)]}\frac{1}{\sigma}\right). \tag{R15}$$

Equation (R15) extends the comparison after main Corollary 25 to each fixed $p > 1$. The logarithmic exponent $p/[2(p-1)]$ tends to $1/2$ as $p \to \infty$; the asymptotic estimates above keep p fixed. Companion C09 reports the high-precision $p = 3$ evaluation separately from the double-precision optimizer record.

D. Modal leading constants

The exact inverse-time modal cost and degenerate cases are given in Technical Supplement Section II-C. For the nondegenerate scaling, fix finitely many modal rates and initial values with $\xi_0 = (\xi_{j0})_j \neq 0$, and set $\sigma_j = \sigma\bar{\sigma}_j$ with fixed $\bar{\sigma}_j \geq 0$ and positive effective noise. Following main Proposition 26, define

$$\rho_* := \min\{\rho_j : \xi_{j0} \neq 0\}, \qquad c_*^2 := \sum_{\rho_j=\rho_*} \xi_{j0}^2 e^{-2\rho_j} T^{-2\rho_j} > 0, \tag{R16}$$

$$S_B := \sum_j \bar{\sigma}_j^2 B_{\rho_j} > 0, \qquad S_b := \sum_j \bar{\sigma}_j^2 b_{\rho_j} > 0. \tag{R17}$$

The coefficients B_{ρ_j}, C_{ρ_j} and b_{ρ_j} are those of main equation (40) evaluated at $\rho = \rho_j$. The active rate ρ_* is the smallest rate present in the initial error, and c_*^2 includes every initially active mode with that rate. The constants S_B and S_b aggregate the noise contributions, including noisy modes whose initial values vanish. The proof in Technical Supplement Section I-E uses the order scale $L_\sigma = \sigma^{-2/(2\rho_*+1)}$. To obtain leading constants, define its constant-normalized version

$$\widehat{L}_\sigma := \left(\frac{2\rho_* c_*^2}{S_B\sigma^2}\right)^{1/(2\rho_*+1)}. \tag{R18}$$

Let $\mathcal{M}_\sigma(K) = \mathbb{E}\|\xi(T)\|^2$, as in the main proposition, and let K_{opt} be any global common-cap minimizer over $K \geq T^{-1}$. The two positive terms in the lower bound proved in Technical Supplement Section I-E, compared at $\widehat{L}_\sigma$, confine $K_{\mathrm{opt}}/\widehat{L}_\sigma$ to a fixed compact subinterval of $(0,\infty)$ as $\sigma \downarrow 0$. Writing $z = K/\widehat{L}_\sigma$, the expansion on that interval is

$$\frac{\mathcal{M}_\sigma(\widehat{L}_\sigma z)}{c_*^2 \widehat{L}_\sigma^{-2\rho_*}} = z^{-2\rho_*} + 2\rho_* z + o(1). \tag{R19}$$

The limit has its unique minimum at $z = 1$. Consequently, every global common-cap minimizer satisfies

$$K_{\mathrm{opt}} \sim \left(\frac{2\rho_* c_*^2}{S_B\sigma^2}\right)^{1/(2\rho_*+1)}, \tag{R20}$$

$$\min_{K\geq T^{-1}} \mathcal{M}_\sigma(K) \sim (2\rho_*+1)(c_*^2)^{1/(2\rho_*+1)} \left(\frac{S_B\sigma^2}{2\rho_*}\right)^{2\rho_*/(2\rho_*+1)}. \tag{R21}$$

These equivalents give the leading constants for main equations (55)–(56), governed by the slowest initially active mode rather than necessarily the slowest graph mode.

E. Bounded deterministic measurement error

The white-noise refinements above use stochastic second moments. For the distinct amplitude-bounded channel in main Proposition 27, let $n(t)$ be a measurable measurement error with $|n(t)| \leq \bar{n}$ and use $\dot{\eta} = -\rho\kappa_K(\eta + n)$, $\eta(0) = \eta_0$. The proposition gives the exact worst-case absolute terminal error, including $\eta_0 = 0$. With $\alpha_{\kappa,K} = e^{-\rho\mathcal{J}_\kappa(K)}$, an admissible profile satisfies $\alpha_{\kappa,K} \to 0$ as $K \to \infty$, and hence

$$E_{\mathrm{wc}}(K) := \alpha_{\kappa,K}|\eta_0| + (1-\alpha_{\kappa,K})\bar{n} = \bar{n} + \alpha_{\kappa,K}(|\eta_0| - \bar{n}) \longrightarrow \bar{n}. \tag{R22}$$

For $|\eta_0| > \bar{n}$, the error approaches $\bar{n}$ from above; for $|\eta_0| < \bar{n}$, it approaches from below; at equality it is constant. The finite-cap error interpolates between $|\eta_0|$ and $\bar{n}$; zero initial error with zero gain gives zero error. This amplitude-bounded criterion complements [33] and differs from white-noise MSE.

II. Secondary Visual Diagnostics

The following illustrations use the same definitions and frozen numerical records as the submitted pair. They complement the complete arguments and primary validation retained there. Figure R1 complements the local certificate in the Technical Supplement with three-graph spectral predictions and late-time fits in (a), and their residuals in (b), for the direct initialization specified in the caption.

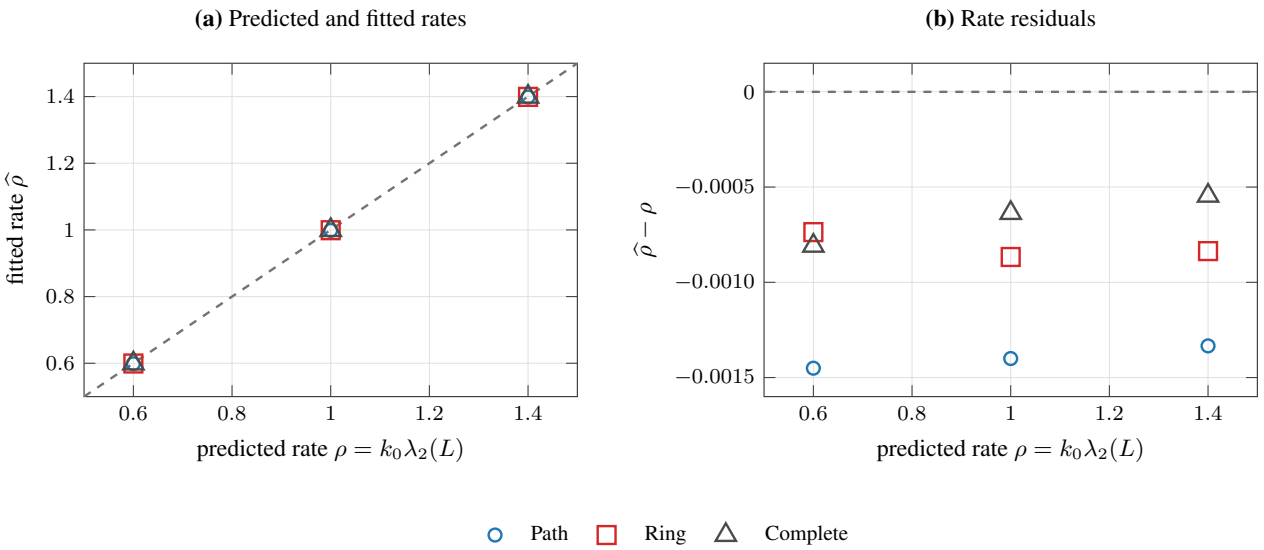


Fig. R1. The deterministic spectral-rate diagnostics use six-node unit-weight path, ring and complete graphs under $\kappa_1(t) = (T-t)^{-1}$, $T = 2$. (a) Nine fitted rates $\widehat{\rho}$ are compared with $\rho = k_0\lambda_2(L)$ and the equality reference. (b) The same rates give signed residuals $\widehat{\rho} - \rho$ around zero. Here $\widehat{\rho}$ is the slope of $\log \|\eta(t)\|_2$ against $\log s$, $s = (T-t)/T$. The finite-amplitude state in Technical Supplement Section IV-A is assigned directly at $t = 1.6$. Each fit uses 272 selected time samples; companion C04, *Nine graph-rate cases*, specifies the sample-selection criteria. Main Proposition 28 and (60) give the spectral prediction.

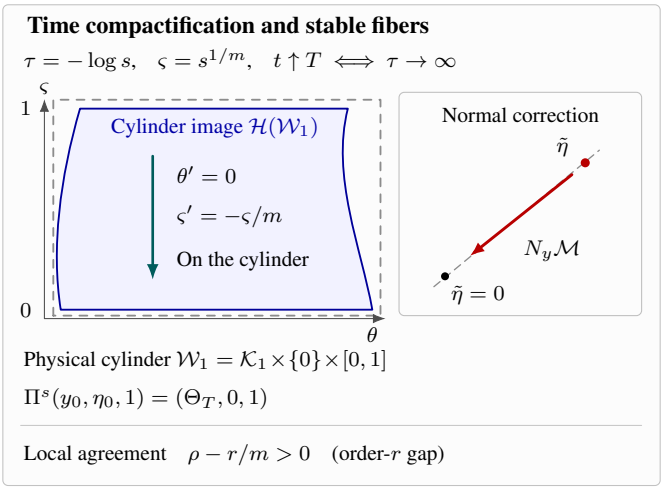


Fig. R2. Inverse-time completion. The physical cylinder has zero-section coordinates $\mathcal{H}(y,\varsigma) = (\theta,\varsigma)$, $\theta = \phi_{T\varsigma^m}(y)$, with prime denoting $d/d\tau$ in main (25). The inset shows instantaneous normal correction at fixed y and ς; $\tilde{\eta}$ is parametrized by τ. Stable leaves are normal-bundle graphs; Π^s assigns the cylinder reference. The completion preserves local equations on the agreement region. Under the collar hypotheses and displayed gap, main Theorem 18 gives extension-compatible persistence; main Corollary 19 gives original-target contact under its terminal condition.